Probability Distribution (1)

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Variable

- A quantity or characteristic that can take different values or vary in different situations.

- For example: Age, Gender etc.

Random variable

- A random variable is a variable
- That can take different values (numerical)
- Determined by the outcome of an experiment

Difference

- The differences between variable and random variable are-

  a) Random variable always takes numerical values

  b) There is a probability associated with each possible values

Example

For example,

A coin is tossed. It has two possible outcomes-Head and Tail.

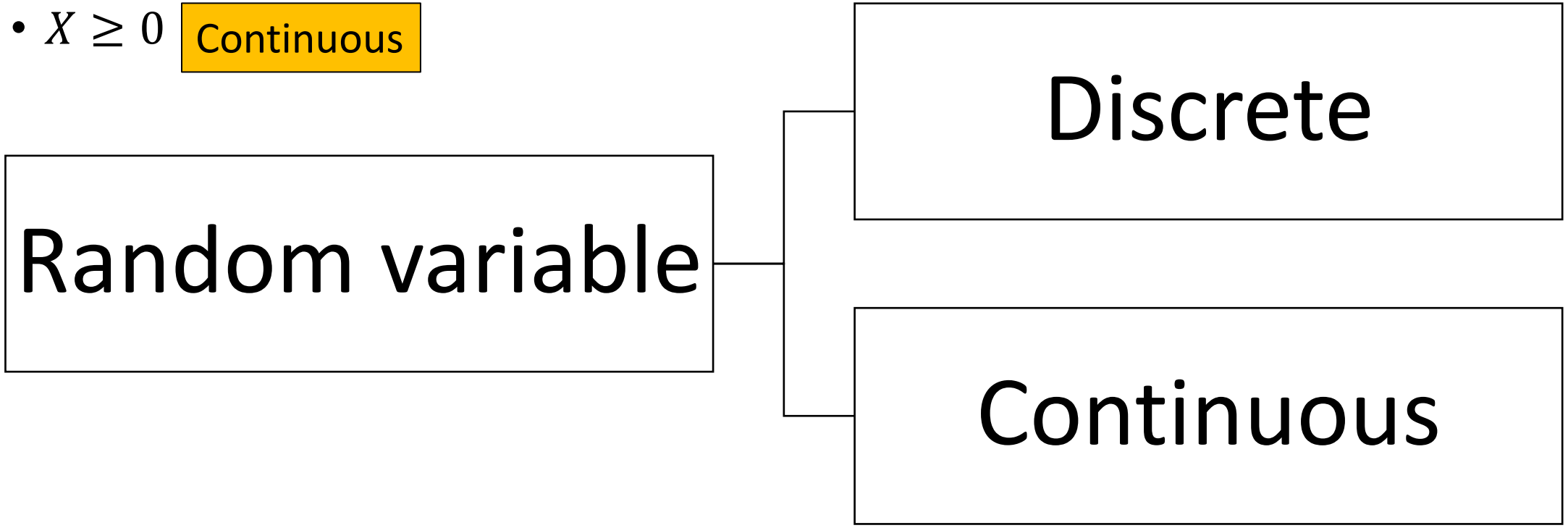
Consider a variable,

$$X = \ outcome\ of\ a\ coin\ toss = \begin{cases} 1; If\ head \\ 0; If\ tail \end{cases}$$

Here, we can write $P(X = 1) = \frac{1}{2}$, $P(X = 0) = \frac{1}{2}$

Types of Random Variable

- $X = \{1, 2, 3, 4, 5, 6\}$ Discrete
- $X \geq 0$ Continuous



Discrete Random Variable

- A random variable is defined over a discrete sample space.

- For example, $X = Number\ of\ student\ in\ a\ particular\ section$

$$\{0, 1, 2, 3, \ldots.\}$$



Discrete random variable

Continuous Random Variable

- A random variable defined over a continuous sample space

- For example, $X = Height\ of\ a\ person$

$$\{0 < X < finite\ number\}$$



Continuous random variable

Examples

Discrete Random Variable:

Continuous Random Variable:

Probability Distribution

- Distribution of the probabilities among the different values of a random variable.

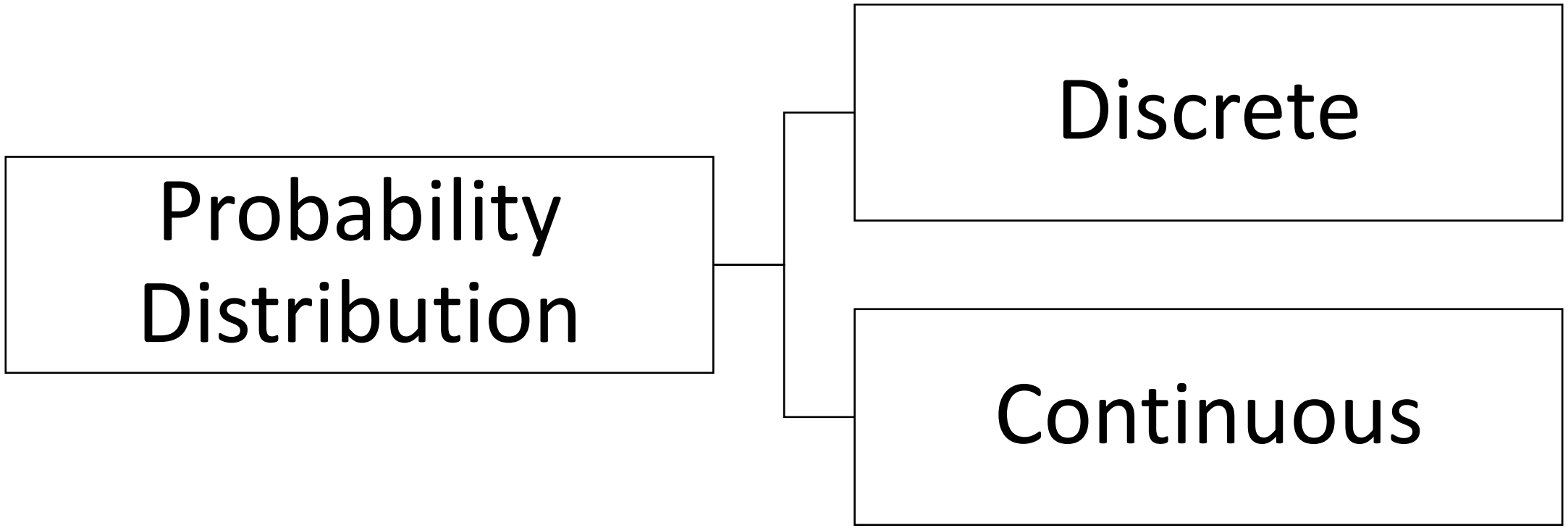
Probability Distribution

- Distribution of the probabilities among the different values of a random variable.

- For example, here's an example of probability distribution that results from the rolling of a single fair die.

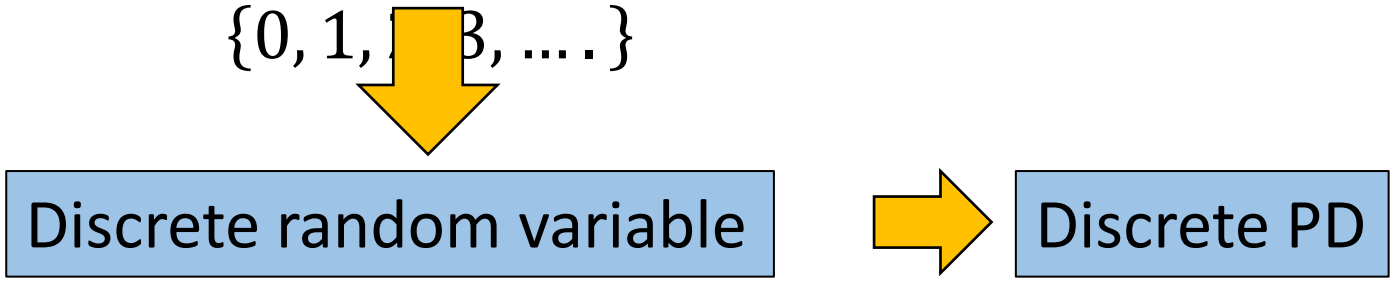
X	**1**	**2**	**3**	**4**	**5**	**6**
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6

Types of Probability Distribution



Discrete Probability Distribution

- Probability distribution of a discrete random variable
- For example, $X = Number\ of\ student\ in\ a\ particular\ section$

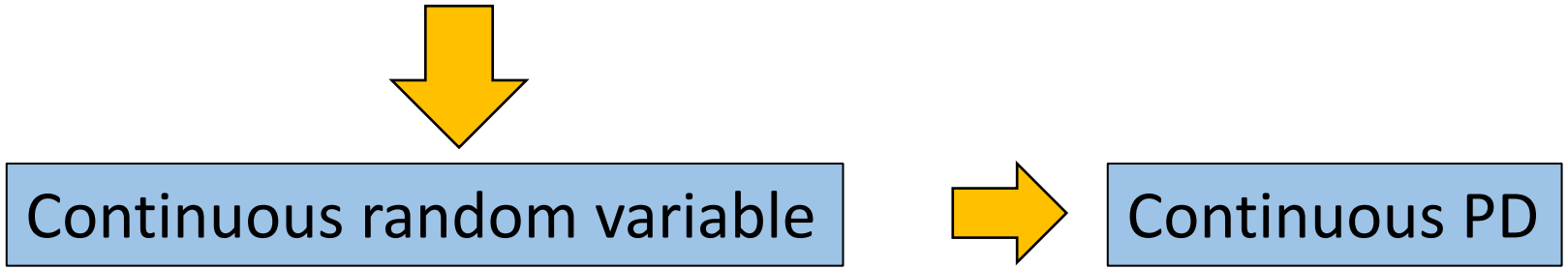


Continuous Probability Distribution

- Probability distribution of a continuous random variable

- For example, $X = Height\ of\ a\ person$

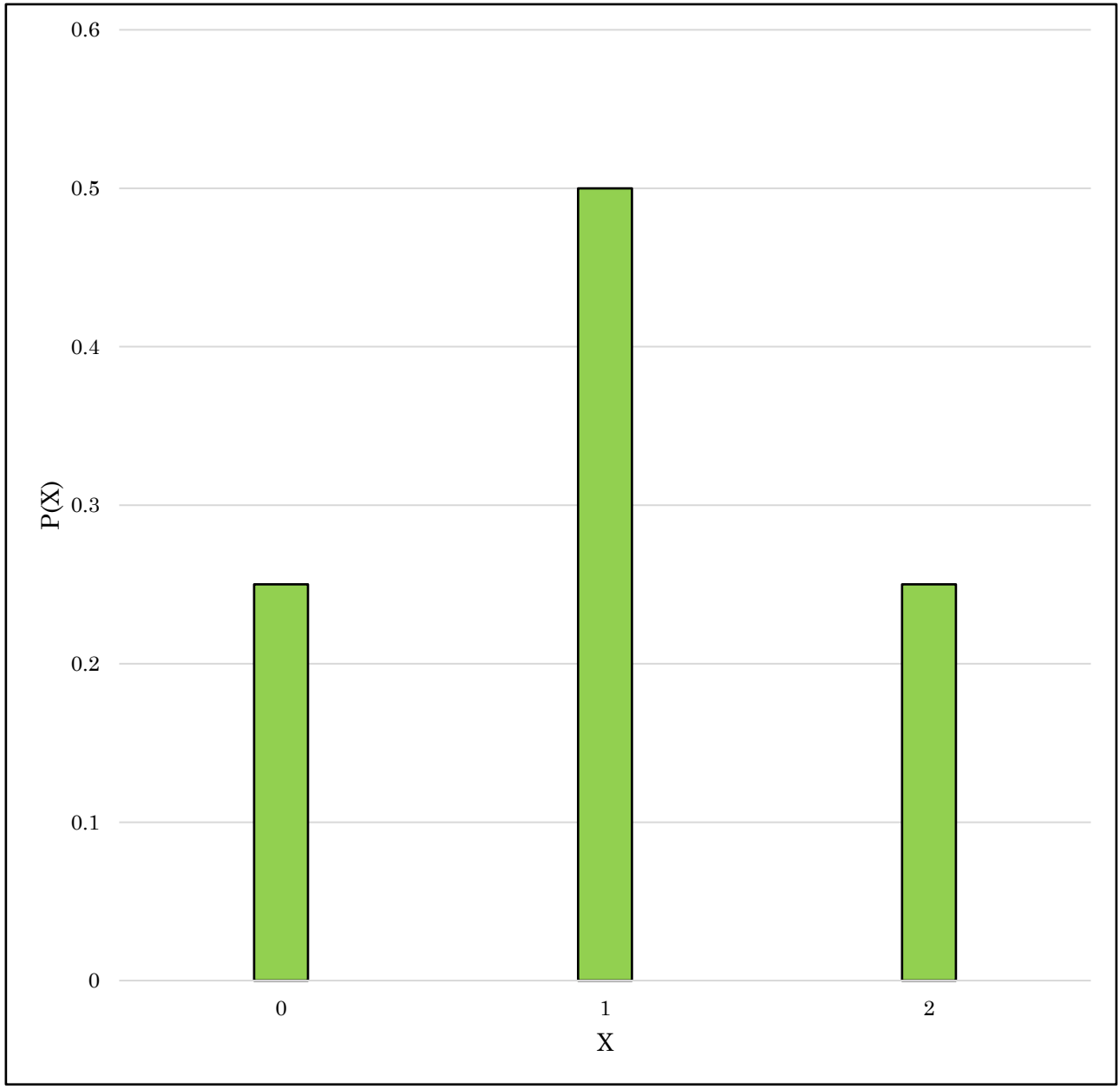
$$\{0 < X < finite\ number\}$$



Example

- Tossing a coin 2 times.
- $X = Number\ of\ tail\ appeared$
- $S = \{HH, HT, TH, TT\}$

X	0	1	2
$P(x)$	1/4	2/4	1/4



Probability Mass Function (PMF)

- The probability distribution function of a discrete random variable X is called a **PMF** and is denoted by $\boldsymbol{P(x)}$

- $\boldsymbol{P(x)}$ satisfy three properties

  a) $0 \leq P(x) \leq 1$
  b) $\sum P(x_i) = 1$
  c) $P(x) = P(X = x)$

$\boldsymbol{X}$	**0**	**1**	**2**
$P(x)$	$1/4$	$2/4$	$1/4$

Probability Density Function (PDF)

- The probability distribution function of a continuous random variable X is called a **PDF** and is denoted by $\boldsymbol{f(x)}$

- $\boldsymbol{f(x)}$ satisfy three properties

$a)\ f(x) \geq 0$

$b)\ \int f(x)\, dx = 1$

$c)\ P(a \leq X \leq b) = \int_a^b f(x)\, dx$

Example

X	0	1	2	3
$P(x)$	$2/25$	$6/25$	$11/25$	$6/25$

$P(X \geq 1)$ ➡ $P(X = 1) + P(X = 2) + P(X = 3)$ ➡ $\frac{6}{25} + \frac{11}{25} + \frac{6}{25} = 0.92$

$P(X \leq 1)$ ➡ $P(X = 0) + P(X = 1)$ ➡ $\frac{2}{25} + \frac{6}{25} = 0.32$

Example

- Suppose you roll a fair six-sided die twice. Let, X denote the number of times you roll a six.

a) Summarize the probability function of X.

b) Determine,

a) $P(X > 0)$ Ans:/36

b) $P(X \leq 1)$ Ans:?/36

c) $P(X = 2)$ Ans:?/36

X	0	1	2
$P(x)$	25/36	12/36	1/36

Mathematical Expectation

- For a discrete random variable X with $PMF\ P(x)$, the mathematical expectation of X is-

$$E(X) = \sum x\, P(x)$$

- For a continuous random variable X with $PDF\ f(x)$, the mathematical expectation of X is-

$$E(X) = \int x\, f(x)\, dx$$

Properties

- $E(X) = Mean\ (X) = \mu(X)$
- $E(X^2) - \big(E(X)\big)^2 = Variance\ (X) = Var(X)$
- $E(c) = c; c\ is\ constant$
- $E(cX) = c\ E(X); c\ is\ constant$
- $E(X + c) = E(X) + c; c\ is\ constant$
- $E(X \pm Y) = E(X) \pm E(Y)$
- $E(XY) = E(X) \times E(Y)$

Properties

- $Var(c) = 0; c\ is\ constant$
- $Var(cX) = c^2\ Var(X)$
- $Var(X + c) = Var(X) + Var(c) = Var(X)$
- $Var(X \pm Y) = Var(X) \pm Var(Y)$

Example

- Find the mean and variance of

X	0	1	2
$P(x)$	1/4	2/4	1/4

$$Mean, E(X) = \sum x\, P(x) = \left(0 \times \frac{1}{4}\right) + \left(1 \times \frac{2}{4}\right) + \left(2 \times \frac{1}{4}\right) = 1$$

$$Var(X) = E(X^2) - \big(E(X)\big)^2 = ???$$

Example

- Individuals applying for a particular country visa are allowed up to four attempts to pass the license exam. An applicant is randomly selected and let X denote the number of attempts made by the applicants. The probability function of X is as follows,

$$P(x) = \begin{cases} \frac{x}{10}; for\ x = 1,2,3,4 \\ 0; Otherwise \end{cases}$$

Find mean of X.

$$E(X) = \sum x \times P(x) = 3$$

The mean number of attempts required to get visa is 3

Example

- Individuals applying for a driving license are allowed up to four attempts to pass the license exam. An applicant is randomly selected and let X denote the number of attempts made by the applicants. The probability function of X is as follows,

$$P(x) = \begin{cases} \frac{x}{10}; for\ x = 1,2,3,4 \\ 0; Otherwise \end{cases}$$

Find mean of X.

$$E(X) = \sum x \times P(x) = 3$$

The mean number of attempts required to pass the license exam is 3

Example

- Let, X be a continuous random variable with pdf,

$$f(x) = \begin{cases} \frac{2x}{3}; & if\ 1 \le x \le 2 \\ 0, & Otherwise \end{cases}$$

Mean

$$E(X) = \int_1^2 x\, f(x) dx$$

$$E(X) = \int_1^2 x \times \frac{2x}{3} dx$$

?????

$$E(X) = \int_1^2 \frac{2x^2}{3} dx$$

Thank You